

Time : 3 Hours

Max. Marks : 60

The candidates shall limit their answers precisely within the answer-book (40 pages) issued to them and no supplementary/continuation sheet will be issued.

Note : Attempt five questions in all by selecting one question each from section A, B, C and D. Section-E is compulsory.

SECTION - A

1. (a) Explain the structure and importance of HTML5 DOM. (6)
(b) Describe different types of CSS with their advantages and disadvantages. (6)
2. (a) What is JavaScript event handling? Explain with an example. (6)
(b) Discuss the role of APIs in web development and their integration. (6)

SECTION - B

3. (a) Explain the Node.js event-driven architecture. (6)
(b) What are Node.js modules? Differentiate between core and local modules. (6)
4. (a) Explain the concept of Middleware in Node.js with an example. (6)
(b) Discuss the significance of the Node.js Package Manager (NPM). (6)

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SECTION - C

5. (a) What is the Node.js file system module? Explain its operations. (6)
(b) Write a Node.js program to read and write a file asynchronously. (6)
6. (a) Explain the debugging process in Node.js using built-in tools. (6)
(b) What is the concept of firmware in web applications? Explain. (6)

SECTION - D

7. (a) Explain the installation and setup of MongoDB. (6)
(b) What is CRUD? Demonstrate CRUD operations in MongoDB with Node.js. (6)
8. (a) Explain the indexing concept in MongoDB and its types. (6)
(b) How does MongoDB handle data replication? Explain. (6)

SECTION - E (Compulsory) (Short Answer Questions)

9. Explain the following:
(a) Define the concept of RESTful APIs and their advantages.
(b) What is the difference between synchronous and asynchronous programming in Node.js?
(c) Explain the use of the Event Emitter in Node.js.
(d) How does NoSQL differ from SQL databases?
(e) Explain the role of MongoDB Compass in database management.
(f) What is the significance of CORS in web applications?